

A choice-free Lean formalization of Bishop–Cheng dominated convergence over presented reals

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Author contributions

The paper has a single author, who conceived the work, carried out the formalization, built and ran the audit apparatus, and wrote the manuscript, and who is responsible for all of it.

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Competing interests

The author has no competing interests to declare that are relevant to the content of this article.

Data and code availability

There is no dataset. The Lean 4 artifact, the audit apparatus and the shipped audit logs are archived and citable at `10.5281/zenodo.22268374`, and are also at <https://github.com/turahigashi/choicefree-bc-dct> under tag `v0.6.1`, the commit the deposit was cut from. Reproducing the build requires Lean 4.30.0 and mathlib 4.30.0; the remaining checks need only Python 3 and `coreutils`.

Ethics approval

Not applicable.

Preprint

A preprint of this work is on arXiv (`math.LO`, cross-listed to `cs.LO`), under CC BY 4.0.